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PEG User's Manual

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For use beginning with
TopAGNPS v6.00.a.000

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PEG User's Manual:

Introduction:

“PEG” (**P**otential **E**phemeral **G**ully) is a module in the suite of modules called TopAGNPS. Its purpose is to allow the user to create an input ephemeral gully data section for use with AnnAGNPS.

Most of the input files for PEG, as described below, can be generated by executing the following modules: DEDNM, RASPRO, RASFOR, and AgFlow. The TopAGNPS modules, (DEDNM, RASPRO, and RASFOR), produces spatial files required by PEG. AgFlow produces spatial output files called “AnnAGNPS_Cell_IDs.asc” and “AnnAGNPS_Reach_IDs.asc” which are also required by PEG.

PEG produces several output files which are described below. The main output file of interest is the “AnnAGNPS_Ephemeral_Gully_Data_Section.csv” output file. This is the file that may be used directly by AnnAGNPS as the “Ephemeral Gully Data” section.

Input files:

There are eight input files, described below, necessary for the successful execution of PEG. The first is a control file and the remaining seven are spatial layer files. The layer files are specifically named as described below.

Control file:

PEG.CSV or PEG.XML

The required control file allows the user to specify a number of parameters that affect the processing of spatial data and the determination of potential ephemeral gully locations. If both “PEG.CSV” and “PEG.XML” are found, the “PEG.CSV” control file will be processed. A message will be produced in the log file and PEG execution terminated if the control file is not present.

“PEG.CSV” is a csv-formatted control file. The first record in the control file is the header record containing at least one of three possible keywords that may be used. The second record is the data record containing the values for the keywords that are specified in the header record. Only two records are read, header and data; therefore, any other information contained within the control file is ignored. The control file must be named “PEG.CSV”.

The following is an Excel example of the csv-formatted control file showing all of the available keywords:

Input	CTI_value	Accum_pct	Display_Drainage_ Option	Display_Location_ Option	ASC_PATH	CSV_PATH
	72		1	1	./Output_Fi les/ASC	./Output_Fi les/CSV

“PEG.XML” is xml-formatted control file. If the xml-format is used, it must adhere strictly to the following –only one tag per line, with no spaces. The xml-formatted control file must be named “PEG.XML”. The following is an example of the xml-formatted control file showing all available keywords.

```
<TOPAGNPS>
<KEYWORD ID="ACCUM_PCT"></KEYWORD>
<KEYWORD ID="ASC_PATH">./Output_Files/ASC</KEYWORD>
<KEYWORD ID="CSV_PATH">./Output_Files/CSV</KEYWORD>
<KEYWORD ID="CTI_VALUE">72.</KEYWORD>
<KEYWORD ID="DISPLAY_DRAINAGE_OPTION">1</KEYWORD>
<KEYWORD ID="DISPLAY_LOCATION_OPTION">1</KEYWORD>
<KEYWORD ID="INPUT"></KEYWORD> </TOPAGNPS>
</TOPAGNPS>
```

Keywords:

All parameters may be present but only required parameters and any desired optional parameters need be present. There are five possible keywords in the control file. All keywords must be spelled exactly as specified below and have a comma separating the keywords. The keywords are case and order insensitive.

“Input”:

This optional keyword allows the user to specify a csv-formatted input file containing PEG point locations to be read in and processed.

If this keyword is included in the control file and contains a valid filename then that file will be opened and the PEG point locations will be read in as described below in the “Input data files” section. The use of either “CTI_value” or “Accum_pct” is optional in this case and may be used as a filter applied to the user-specified inputted PEG points. If neither, the “CTI_value” nor the “Accum_pct” keyword is used then all PEG points in the user input file will be considered and flowpaths checked for validity.

If this keyword is included in the control file but its value is left blank then it is ignored and PEG points will be internally defined. If blank, then one of either “CTI_value” or “Accum_pct” is required with a valid value.

NOTE: The output files "PEG_Summary.csv" and "PEG_Summary.txt" are reserved and cannot be used as the PEG points input filename. "PEG_Summary.csv" can be renamed and then used directly as input back into the PEG program though only the x & y coordinates are read and used from this file; no other information is retained.

"CTI_value":

This keyword allows the user to set a threshold based on a CTI value.

This keyword is optional if the "Input" keyword is present and contains a valid filename. Otherwise, one of the "CTI_value" or "Accum_pct" keywords must be present and contain a value. Both, "CTI_value" and "Accum_pct" keywords may be included in the control file but only one may contain a value and the other left blank.

"Accum_pct":

This keyword allows the user to set a threshold based on a percent value.

This keyword is optional if the "Input" keyword is present and contains a valid filename. Otherwise, one of the "CTI_value" or "Accum_pct" keywords must be present and contain a value. Both, "CTI_value" and "Accum_pct" keywords may be included in the control file but only one may contain a value and the other left blank.

"Display_Drainage_Option":

This keyword allows the user to select how drainage area to the PEG points are displayed.

This keyword is optional. The acceptable values are "0" and "1". A value of "0" indicates that the display of peg point drainage areas may overlap other peg points and their associated drainage areas. A value of "1" indicates that the display of peg point drainage areas may not overlap other peg points and their associated drainage areas.

Options: 0 or 1; DEFAULT = 0

This option along with the "Display_Location_Option" option determines the content of the spatial layer output file called "PEG_Drainage_Area_to_Valid_Peg_Points.asc".

"Display_Location_Option":

This keyword allows the user to select which PEG point drainage areas are to be displayed.

This keyword is optional. The acceptable values are "0", "1", or "2". A value of "0" indicates that all valid peg point drainage areas will be displayed. A value of "1" indicates that only valid cell-located peg point drainage areas will be displayed. A value of "2" indicates that only valid reach-located peg point drainage areas will be displayed.

Options: 0, 1, or 2; DEFAULT = 0.

This option along with the "Display_Drainage_Option" option determines the content of the spatial layer output file called "PEG_Drainage_Area_to_Valid_Peg_Points.asc".

"ASC_Path":

This optional keyword allows the user to specify the path to where the spatial files created by DEDNM, RASPRO, RASFOR, and AgFlow are located. This path will also be used for spatial output files created by PEG.

This keyword is optional and if not included then the default location of the spatial input files and PEG spatial output files will be assumed. The default relative path is the folder where TopAGNPS was invoked. If the keyword is present with a path specified then that path will be used for all of the required spatial input files and PEG spatial output files. If the keyword is used but its value is blank, then the current path from which TopAGNPS is executed will be used for all of the spatial input files and PEG spatial output files.

NOTE: This keyword is only used for independent runs of the PEG module. It is ignored when PEG is run in sequence with DEDNM, RASPRO, RASFOR, and AgFlow. In this case the output files from PEG will be placed in the default folder location.

"CSV_Path":

This optional keyword allows the user to specify the path to where the csv-formatted output files created by PEG will be written.

This keyword is optional and if not included then the default location of the PEG csv-formatted output files will be assumed. The default relative path is the folder where TopAGNPS was invoked. If the keyword is present with a path specified then that path will be used for all of the PEG csv-formatted output files. If the keyword is used but its value is blank, then the current path from which TopAGNPS is executed will be used for all of the PEG csv-formatted output files.

NOTE: This keyword is only used for independent runs of the PEG module. It is ignored when PEG is run in sequence with DEDNM, RASPRO, RASFOR, and AgFlow. In this case the output files from PEG will be placed in the default folder location.

Input data files:

If executing TopAGNPS on a UNIX/Linux system (case sensitive), the following list is in the expected case for each file.

(Required):

The following input data files described below are spatial layers. The spatial layers must have the same dimensions with the same header information in each file. All of the filenames must be as specified below.

BOUND.ASC (TopAGNPS):

This required spatial input file is the watershed boundary layer.

FLOVEC.ASC (TopAGNPS):

This required spatial input file is the flow vector layer.

NETW.ASC (TopAGNPS):

This required spatial input file is the stream network layer.

RELIEF.ASC (TopAGNPS):

This required spatial input file is the relief layer.

AnnAGNPS_Cell_IDs.asc (AgFlow):

This required spatial input file is the subwatershed (AGNPS cell) designations layer.

AnnAGNPS_Reach_IDs.asc (AgFlow):

This required spatial input file is the reach (AGNPS reach) designations layer.

TSLOPE.ASC (TopAGNPS):

This required spatial input file is the slope layer.

UPAREA.ASC (TopAGNPS):

This required spatial input file is the drainage area layer.

The following input data file described below is the csv-formatted cell data section file. All of the filenames must be as specified below.

AnnAGNPS_Cell_Data_Section.csv (AgFlow):

This required csv-formatted input file is the subwatershed (AGNPS cell) designations layer.

(Optional - user specified):

PEG allows the user to specify an input file with predetermined peg points if desired. If the user desires to provide this file, then it must conform to the following csv format:

The first line is a required header line although it is ignored by PEG.

Starting with line 2, every line must have at least two data values, separated by a comma “,”, representing an “x” and “y” coordinate location that falls within the bounds of the related spatial layers as described below. PEG will then calculate a related row and column coordinate that matches the row and column coordinates within the spatial layers. More data values may be present on any line but will be ignored.

If the user does not request that an input file is used, then PEG will calculate all cell-located and reach-located peg points. Two summary files produced by PEG called “PEG_Summary.csv” and “PEG_Summary.txt” could then be edited and used as the input file back into PEG.

PEG examines each peg point in the input file and determines if it is a valid cell-located or reach-located peg point and results will be reported accordingly. Otherwise, it will be ignored.

Output files:

There are currently eight output files generated by the program.

Log file:

The log file, “PEG.log”, is automatically generated and is a log of the processing as the program continues its execution. Additionally, the log files for “TopAGNPS” will also contain information specific for PEG. The log file, as it is being written, is also displayed to the screen. The log file contains processing information as well as error and warning messages.

Intermediate Output data files:

PEG_cti_cell_table.txt:

This is a computed intermediate file containing information about potential ephemeral gullies located in the AGNPS cells.

Note: The column labeled "CTINDEX" will contain "-9999.0000" if neither "CTI_value" nor "Accum_pct" keywords were used in "PEG.inp".

PEG_cti_grid_table.asc:

This is a computed intermediate spatial layer calculated by (planform curvature value * slope * drainage area) for each raster.

Note: This file is not produced if neither "CTI_value" nor "Accum_pct" keywords were used in "PEG.inp".

PEG_cti_reach_table.txt:

This is a computed intermediate file containing information about potential ephemeral gullies located in the AGNPS reaches.

Note: The column labeled "CTINDEX" will contain "-9999.0000" if neither "CTI_value" nor "Accum_pct" keywords were used in "PEG.inp".

PEG_cti_summ_table.txt:

This is a computed intermediate file containing CTI information.

Note: This file is not produced if neither "CTI_value" nor "Accum_pct" keywords were used in "PEG.inp".

PEG_points.asc:

This is a computed intermediate spatial layer containing valid PEG point locations.

PEG_Drainage_Area_to_Valid_Peg_Points.asc:

This is a computed intermediate spatial layer containing the drainage areas to each valid PEG point location. This file is created based on the values used for the “Display_Drainage_Option” and “Display_Location_Option” keyword options listed above.

Output data files:**PEG_Summary.csv & PEG_Summary.txt:**

These output files contain expanded information about each potential ephemeral gully in the cells and reaches. One is in csv format and the other in a fixed-format text file.

Note: The column labeled “CTINDEX” will contain “-9999.0000” if neither “CTI_value” nor “Accum_pct” keywords were used in “PEG.inp”.

AnnAGNPS_Ephemeral_Gully_Data_Section.csv:

This output file is a csv file that contains information that can be imported as the “Ephemeral Gully Data” section into the AGNPS Input Editor (v5.41+) or used directly by AnnAGNPS (v5.41+) if used with the AnnAGNPS csv input option.

PEG_Peg_Points_Not_Used_For_AnnAGNPS.csv:

This output file is a csv file that contains information PEG points that were not selected as valid AnnAGNPS PEG point locations. There are a number of reasons that a PEG point might be considered as invalid. Below is a list of reason codes that are included in this output file:

- 1 = Drainage area to mouth of gully is too small. AnnAGNPS minimum area is .00001 [ha] or .1 [m2]
- 2 = Calculated slope is greater than AnnAGNPS maximum of 3.0 [nd]
- 3 = Invalid cell-located peg point. While its threshold value was met; its immediate adjacent downstream raster does not fall on a reach.
- 4 = Invalid cell-located peg point. While its threshold value was met; its immediate adjacent downstream raster also meets the threshold value and is located within the same cell.
- 5 = Invalid cell-located peg point because its calculated cti value was negative. Negative cti values are not allowed.
- 6 = Invalid cell-located peg point because its cti value does not exceed the threshold value.
- 7 = Invalid reach-located peg point because its cti value does not exceed the threshold value.
- 8 = Invalid peg point because it is located outside of the watershed boundary. Applies when the optional PEG point input file is used.

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9 = Invalid reach-located peg point because it's drainage area is larger than the local cell's drainage area to the downstream end of the reach.